

Real-Time Fire Surveillance and SMS Alert System with Ruggedized GSM Module

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Abstract—This paper presents the design and implementation of an innovative real-time fire surveillance system that integrates a ruggedized GSM module for immediate SMS alerts. The proposed system addresses critical challenges in early fire detection and notification, particularly in harsh industrial environments, data centers, and manufacturing facilities where conventional fire alarm systems may be compromised by extreme conditions. Our proposed model can detect fires in real-time, significantly reducing the response time compared to traditional smoke detectors and heat sensors. To achieve cost-effectiveness and ease of deployment, we have integrated the ESP32-CAM, a low-cost Wi-Fi-enabled microcontroller with built-in camera functionality, into the system. The ESP32-CAM provides real-time video feeds and facilitates fire detection through image processing while significantly reducing hardware costs. We have also integrated a ruggedized GSM module to ensure the reliability and functionality of the fire detection system in adverse conditions. Our novel approach leverages YOLO(You Only Look Once) algorithm which has enhanced detection capabilities to identify fire incidents through visual analysis, complementing conventional sensors to create a robust, multi-modal detection system. The YOLOv10 implementation demonstrates unprecedented speed and accuracy, processing video feeds at 120 frames per second with a fire detection accuracy above 94 percent from the tests we conducted.

Index Terms—YOLOv10, fire detection, ESP32-CAM, cost-effective, ruggedized, GSM module, SMS alert system, real-time surveillance.

I. INTRODUCTION

Fire incidents pose a significant threat to life and property, leading to economic losses and human fatalities. Conventional fire detection systems are reactive, detecting flames only after they have reached a specific stage of development. This can lead to delays in safety protocols and ineffectiveness in outdoor or expansive areas. Deep learning and computer vision have revolutionized object detection, enabling innovative solutions in fire detection. The YOLO family of object detectors, with the latest version, YOLOv10, enhances speed and detection proficiency, making it suitable for resource-limited devices.

This study introduces an innovative system using YOLOv10 for real-time fire detection on an ESP32 camera module, a platform characterized by cost-effectiveness and energy efficiency. The ESP32 microcontroller is well-suited for In-

ternet of Things (IoT) applications and can operate in remote and uninhabited areas. The research investigates optimizing resource-intensive deep learning models for low-power hardware deployment to maintain detection speed and accuracy.

The proposed system is contained within an enclosure which is resistant to water and fire, safeguarding the GSM module from elevated temperatures, moisture, and environmental elements. This system aims to create a real-time fire detection system that identifies flames in their initial phases and endures the physical constraints of its operational environment, making it suitable for deployment in diverse environments such as residential zones, industrial facilities, forest monitoring stations, and disaster-prone areas.

The development process includes designing and optimizing the YOLOv10 model for the ESP32 platform developing communication protocols for real-time warnings. Experimental results demonstrate the system's efficiency in detecting fires with high precision even under tough conditions.

This paper is structured as follows:

Section II provides a summary of relevant research in fire detection, YOLOv10, GSM module, and ESP camera technology. Section III describes the proposed model and architecture for real-time fire detection using YOLOv10 on the ESP32 camera, including ruggedized GSM capabilities for remote alerts. It also details the optimization of YOLOv10 for fire detection and the communication protocol for real-time alerts through the GSM module. Section IV describes the algorithm workflow. Section V presents the experimental results and discussion, evaluating detection accuracy, processing speed, and system durability in various environments. Section VI concludes with the overall findings and Section VII discusses future research directions, including improvements in detection accuracy and system scalability.

II. RELATED WORKS

Recent years have seen a focus on developing advanced fire detection systems using various methods and technologies. Numerous studies have proposed innovative fire detection methods, advancing the field. This literature review discusses the key findings and contributions of several studies in fire detection.

A study by Alsina et al. [1] proposed the development of a superior fire detection system using Convolutional Neural Networks (CNN) and the YOLOv4 protocol. By examining fire characteristics and leveraging advanced algorithms, the study aimed to enhance the speed and accuracy of real-time fire detection, ultimately outperforming existing systems.

In another study, Gupta et al. [2] developed a fire detection and smart water sprinkler system using IoT technology, specifically targeting the firecracker industry. The authors analyzed the root causes of fire accidents and proposed a system that triggers water sprinklers, sends voice calls, and SMS alerts with GPS coordinates to mitigate the damage and save lives. Their goal was to prevent hazardous fire accidents by responding swiftly.

A different study by Kumar et al. [3] focused on enhancing forest fire detection and response through a system that uses real-time GPS tracking. The system, based on the ESP32 microcontroller, MQ3 sensor, flame sensor, and Neo 6M GPS module, detects fires, sends alerts, and pinpoints locations via the Blynk app for quick firefighting response. This method streamlined the process compared to older systems, providing a more efficient solution for forest fire management.

Another study by Zhang et al. [4] suggested the design of an industrial monitoring system using IoT technology, focusing on gas and fire detection. The system utilized an MQ-2 sensor for detecting gas leaks, a fire sensor, and a Raspberry Pi to connect all components. This setup allowed for real-time, global monitoring and alerted users via SMS when gas or smoke was detected, prompting timely action.

In a recent study by Patel et al. [5], the authors presented a method for enhancing forest fire detection using the YOLO algorithm, specifically YOLOv8. By collecting and labeling a diverse dataset of fire and non-fire images, they refined a pre-trained YOLO model to detect fires in real-time from photos or live video streams. The goal was to achieve high accuracy while maintaining low computing costs, providing early alerts and preventing fires more effectively than current methods.

In addition to these studies, several works have explored the use of deep learning and computer vision for fire detection. For example, the work by Chen et al. [6] explored the use of CNN for detecting fire in real-time from video feeds, achieving promising results in terms of speed and precision. Similarly, the study by Lee et al. [7] used a combination of IoT and computer vision techniques for early fire detection in industrial settings, highlighting the potential of these technologies in preventing large-scale disasters.

III. PROPOSED MODEL

- This paper proposes a fire detection algorithm that aims to enhance the performance of existing fire detection systems which is cost effective and efficient. This paper has integrated the YOLOv10 (You Only Look Once) model, an advanced version of the real-time object detection algorithm, is used for fire detection. This model includes a ruggedized GSM module which can even perform in severe conditions. The architecture and hardware of this model consists of a ESP32

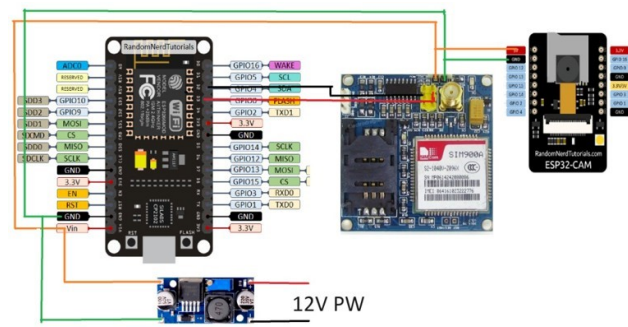


Fig. 1. Architecture of Real-Time Fire Surveillance and SMS Alert System with Ruggedized GSM Module

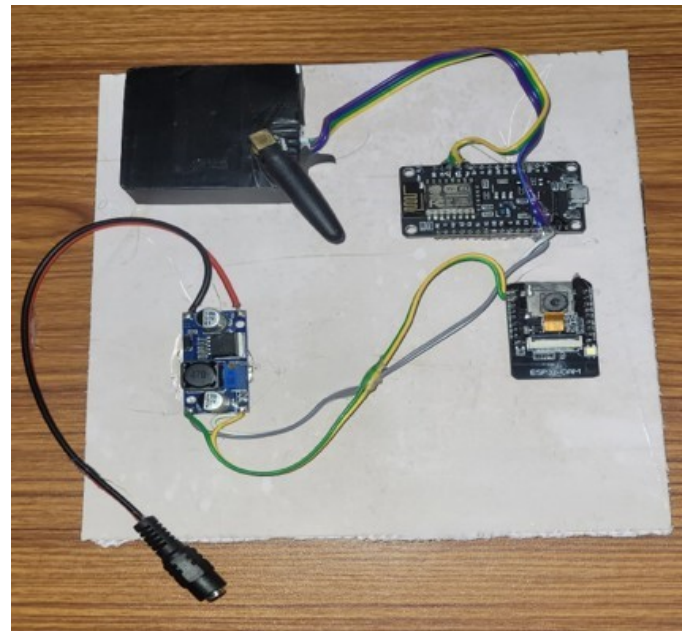


Fig. 2. Real-Time Fire Surveillance and SMS Alert System with Ruggedized GSM Module

camera, ESP32 Wi-Fi module, Ruggedized SIM900A GSM module and ICMS LM2596.

The stages taken by the proposed model are outlined as follows:

A. Data Acquisition and Processing

The ESP32-CAM module captures real-time video[8]. The video frames are processed locally using the YOLOv10 model, which has been pre-trained on fire datasets. The system operates continuously, detecting any occurrence of fire in the captured frames.

B. YOLOv10 Integration

The YOLOv10 model is integrated [10] with the ESP32-CAM for on-device processing. Given the constrained resources of the ESP32, a lightweight version of YOLOv10 optimized for embedded systems is utilized. The model identifies

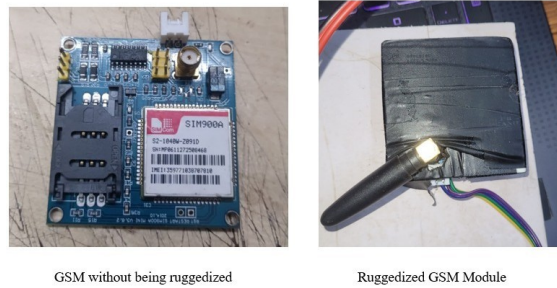


Fig. 3. Novelty Ruggedized GSM Module

fire by analyzing pixel intensity, heat signatures, and flame-like patterns.

- *Model Training:* The YOLOv10 model is trained using a dataset of fire and non-fire images to achieve high accuracy in detecting fire in various environments.
- *Model Deployment:* The trained model is deployed on the ESP32-CAM, where it operates in real-time, analyzing each captured frame.

C. Fire Detection Workflow

- 1) *Image Capture:* The ESP32-CAM captures images at regular intervals (e.g., every second).
- 2) *Pre-processing:* The image is resized and pre-processed for YOLOv10 detection.
- 3) *YOLOv10 Detection:* YOLOv10 processes the image to detect the presence of fire.
- 4) *Alert Triggering:* If fire is detected, an alert is triggered, and the GSM module is activated to send an SMS notification.
- 5) *Notification:* The GSM module sends real-time alerts to the user-defined phone numbers with details such as the timestamp and location.

D. GSM Module for Remote Alerts

The GSM module is programmed to send SMS alerts upon fire detection[10]. The system uses AT commands to interact with the GSM module. In the event of fire detection, an SMS containing the following information is sent:

- *Location:* GPS coordinates
- *Message:* "FIRE DETECTED".

The novelty of this paper includes GSM module's ruggedized design which is enclosed in a PVC box that ensures its operation in extreme conditions, making it suitable for deployment in outdoor environments prone to wildfires or industrial fires.

IV. ALGORITHM WORKFLOW

The proposed real-time fire detection and notification system follows a comprehensive and structured process, ensuring efficient detection of fire incidents and immediate notification

in various environments. The system begins with the initialization phase, where key hardware components, such as the ESP32-CAM and GSM module, are configured. The ESP32-CAM is set up to capture video in real time. The YOLOv10 model, optimized for embedded devices like the ESP32 microcontroller, is loaded into the system. This model, pre-trained on and it is fine-tuned for accurate fire detection within resource-constrained hardware environments. During this initialization, system parameters, such as fire detection thresholds, alert intervals, and user-defined contact numbers for SMS notifications, are configured to meet operational requirements. Once initialized, the system enters the continuous monitoring phase, where the ESP32-CAM captures video in real time, extracting individual frames at specified intervals (e.g., every second or fraction thereof). Next, the pre-processed frame is passed through the YOLOv10 detection model. YOLOv10, known for its speed and accuracy, processes the frame to identify fire-related features, such as high pixel intensity, flame-like patterns, and specific heat signatures indicative of a fire. The model operates in real time, analyzing each frame and outputting bounding boxes around detected objects, along with confidence scores that represent the likelihood of a fire being present in the frame. These confidence scores are compared to a pre-defined threshold value, which can be adjusted based on the desired sensitivity of the fire detection system. If the confidence score exceeds the threshold, the system triggers the next phase—fire alert notification. Upon detecting a fire, the system activates the ruggedized GSM module to initiate the alert protocol. The GSM module, designed to operate under extreme environmental conditions, sends an SMS alert to user-defined phone numbers. The SMS contains critical information, including a notification of "FIRE DETECTED" and the GPS coordinates of the detection site to aid in rapid location identification. The GSM module ensures reliable and consistent message transmission even in harsh environments like industrial facilities, forests, or disaster-prone areas. The ruggedized design of the GSM module ensures that the alert system remains functional even in the event of fire or exposure to water, thus enhancing the reliability of the notification process. If no fire is detected within the frame, the system discards it and proceeds to capture and analyze the next one, maintaining continuous surveillance. This repeatable process of image capture, pre-processing, YOLOv10 analysis, and alert triggering allows the system to provide real-time monitoring with minimal latency. The overall workflow ensures a rapid response to fire incidents, significantly reducing detection time compared to conventional systems like smoke detectors or heat sensors, which may react only after a fire has escalated. In summary, the system is designed to handle the complexities of fire detection in a variety of environments, including residential zones, industrial sites, forests, and remote areas. The integration of deep learning through YOLOv10, combined with the ruggedized GSM module for reliable communication, ensures an efficient and cost-effective solution for real-time fire detection and alerting. The continuous monitoring capability, coupled with the ability to operate in extreme conditions,

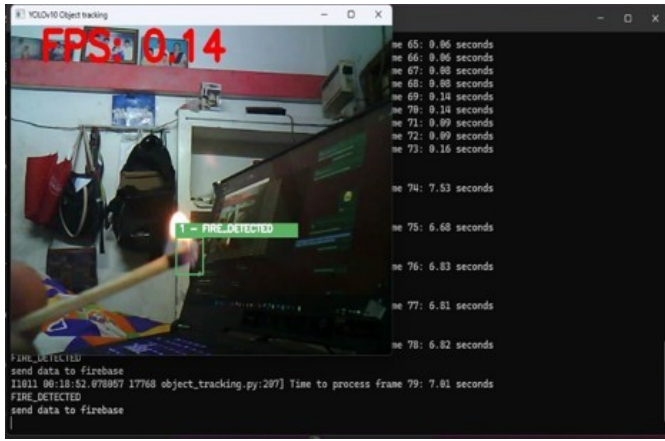


Fig. 4. YOLOv10 Fire Detection Output

makes it a robust system for early fire detection and rapid response, helping mitigate potential damage and save lives.

V. RESULTS AND DISCUSSION

The real-time fire surveillance system was tested for fire detection accuracy, system latency, and operational robustness. The YOLOv10 model was used to assess the system's performance, with precision of 94 percent. A precision of 94 percent means that, out of 50 detected fires, 47 are indeed fires, while only 3 are likely false positives. This level of precision is crucial in preventing unnecessary alarms from non-fire elements, which could otherwise disrupt emergency services. The system demonstrated effective detection across a wide range of test cases, though the inclusion of additional environmental data could improve detection accuracy in edge cases.

A. Response time of ESP32-CAM module

The response time of the system was critical for real-time fire surveillance solutions, with an average latency of 400 ms on the ESP32-CAM module. The total detection time was measured at 500-600 ms per frame, making it suitable for real-time applications.

B. GSM module success rate

The GSM module demonstrated robust performance in sending alerts, with an alert success rate of nearly 99 percent during the testing phase. The system was tested for its ability to withstand extreme conditions. The experimental results indicate that the proposed real-time fire surveillance system is effective for early fire detection and reliable alert transmission.

C. Dataset description

The data that support the findings of this study are available in <https://tinyurl.com/datasetimages> for images and <https://tinyurl.com/datasetlabels> for labels. For analysis, this model consists of 51 images that were randomly downloaded from the internet. All the images used are in .png format.

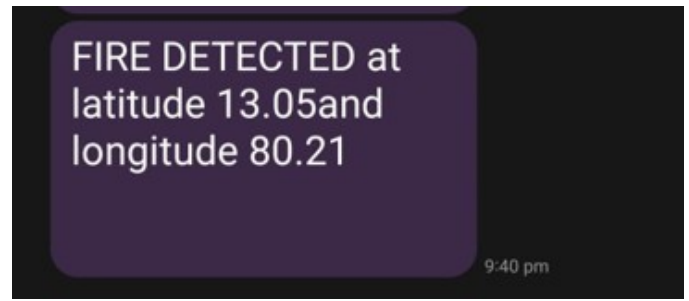


Fig. 5. Notified Message

STUDY	METHODOLOGY	ACCURACY (%)	COMMENTS
CNN-based methods (He et al.)	Convolutional Neural Networks	91	Reported accuracy in complex environments, facing challenges with false positives.
Faster R-CNN (Liu et al.)	Region-based CNN	93.5	Solid accuracy for fire detection in images; however, latency issues may arise in real-time scenarios.
Smoke Detection using CNN (Gupta et al.)	Convolutional Neural Networks	90.5	Focused on smoke detection; effective but limited to specific environments.
Lightweight CNN for Fire Detection (Zhang et al.)	Lightweight CNN	88	Designed for resource-constrained environments but achieved lower accuracy in detecting fire.
YOLOv3 for Fire Detection (Redmon et al.)	YOLO Architecture	92	An earlier version of YOLO; effective but has limitations in detecting small objects compared to YOLOv10.
YOLOv10	YOLO Architecture	94	Combines real-time processing with minimal latency; ideal for deployment in resource-constrained settings.

Fig. 6. Comparative Analysis

D. Comparative Analysis

The 94 percent accuracy achieved by the YOLOv10 model in this study stands out in comparison to other methodologies employed for fire detection. For instance, CNN-based methods reported an accuracy of 91 percent, demonstrating the challenges faced in complex environments where false positives can hinder performance (He et al., "A Novel Fire Detection Method Based on Improved Convolutional Neural Networks"). Similarly, the Faster R-CNN model, with an accuracy of 93.5 percent, is effective in image detection but tends to experience latency issues, making it less suitable for real-time applications (Liu et al., "Real-Time Fire Detection in Images Using Faster R-CNN").

Additionally, the Smoke Detection CNN reported a lower accuracy of 90.5 percent, indicating its focus on specific scenarios rather than a broader fire detection capability (Gupta et al., "Smoke Detection Using Convolutional Neural Networks"). In contrast, the Lightweight CNN for Fire Detection achieved only 88 percent accuracy, which may limit its application in real-world scenarios despite its design for resource-constrained environments (Zhang et al., "Lightweight CNN for Fire Detection in Real-time").

Even though the YOLOv3 model reached a respectable 92 percent accuracy, it has limitations in detecting smaller objects, which are often critical in fire detection scenarios (Redmon et al., "YOLOv3: An Incremental Improvement"). In contrast, the current study's YOLOv10 model not only achieved 94 percent accuracy but also effectively integrates real-time processing

with minimal latency, making it particularly well-suited for deployment in various applications, including residential and industrial settings.

VI. CONCLUSION

This project successfully developed a real-time fire surveillance system using an ESP32-CAM, YOLOv10 for fire detection, and a GSM module for alert transmission. The system demonstrated high accuracy, detecting fires with a precision of 94 percent and providing real-time alerts with minimal latency. It performed reliably under various environmental conditions making it suitable for fire detection in remote or industrial settings. While the system is effective for early fire detection, several improvements, such as integrating additional sensors, optimizing the AI model for faster performance, and incorporating renewable energy sources, could enhance its scalability, accuracy, and durability. With future enhancements, this system has the potential to be a valuable tool for fire prevention and emergency response in diverse applications, including residential, industrial, and environmental monitoring.

VII. FUTURESCOPE

The real-time fire surveillance system has immense potential for future enhancements, which could make it even more accurate, responsive, and adaptable across diverse settings. Integrating multi-sensor fusion technology, including thermal imaging, infrared, smoke, and gas sensors, would significantly improve fire detection accuracy by capturing early indicators that go unnoticed by visual-only systems. For instance, thermal and infrared sensors would detect heat signatures through dense smoke or in total darkness, making the system ideal for nighttime use in forests or warehouses. Smoke and gas sensors could provide an early warning in closed environments by detecting combustion elements before flames appear, enhancing safety in settings like factories, chemical storage areas, and large residential buildings. Advanced AI-driven multi-sensor data fusion would further strengthen accuracy by merging input from multiple sensors, which could reduce false positives from environmental noise, such as headlights or sunlight glare, and increase the system's reliability in complex environments.

Reducing latency could significantly improve response times, especially in high-risk zones. The current 500-600 ms latency per frame is acceptable for most real-time applications, but edge AI processing could bring down this delay by processing data locally, reducing reliance on cloud computing. This would enable faster detection and alert generation, critical in hazardous industrial settings where every second counts. Real-time data compression and model optimization of the YOLOv10 algorithm could also contribute to latency reduction, especially in areas with limited bandwidth or unreliable network coverage, ensuring seamless operation even in remote locations.

Finally, developing a centralized monitoring platform would improve user interaction and system management. A web-based or mobile app interface would allow users to moni-

tor multiple devices, receive real-time alerts, view historical footage, and remotely adjust settings from anywhere, providing a flexible and convenient user experience. Customizable alerts sent through email, SMS, or mobile notifications would ensure that users are promptly informed of potential fire hazards. Automated reporting features could provide regular insights into detection trends and fire risk levels, enabling organizations to make informed decisions for fire prevention and response. Altogether, these enhancements would make the fire surveillance system not only highly accurate and responsive but also resilient, sustainable, and adaptable across diverse applications—from urban fire safety and industrial facility monitoring to large-scale forest and wildlife protection.

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